

BEE 208 — Question 1: Power Factor Status Checker

Control Statement: if...else if...else | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variable pf (double)

Step 3: Prompt user to enter the power factor

Step 4: Read pf

Step 5: If pf < 0 OR pf > 1, display 'Invalid power factor'

Step 6: Else if pf >= 0.00 AND pf <= 0.50, display 'Poor power factor'

Step 7: Else if pf >= 0.51 AND pf <= 0.80, display 'Fair power factor'

Step 8: Else if pf >= 0.81 AND pf <= 0.95, display 'Good power factor'

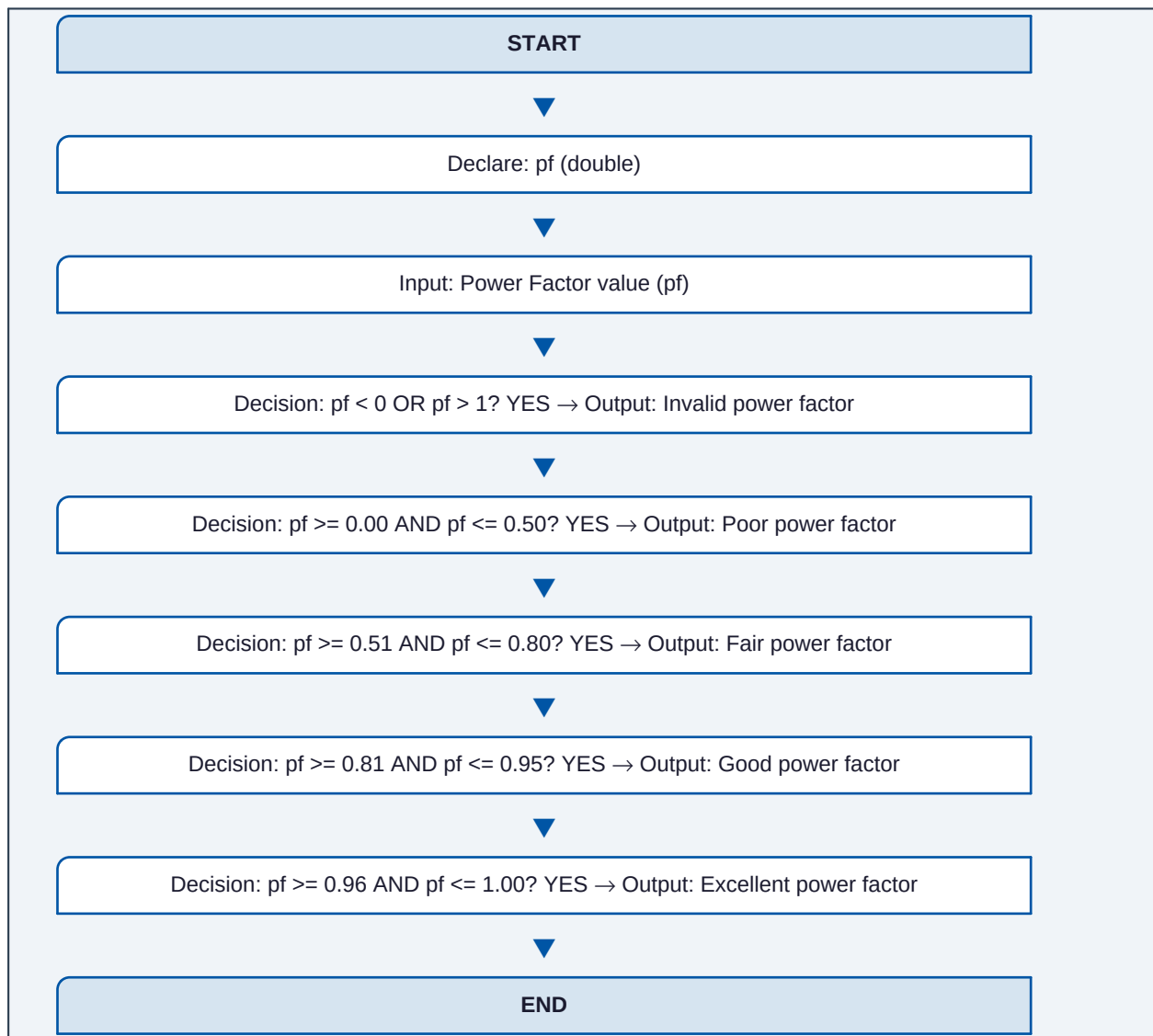
Step 9: Else if pf >= 0.96 AND pf <= 1.00, display 'Excellent power factor'

Step 10: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE pf AS DOUBLE  
  
OUTPUT 'Enter Power Factor (0 to 1): '  
  
INPUT pf  
  
IF pf < 0 OR pf > 1 THEN  
    OUTPUT 'Invalid power factor'  
ELSE IF pf >= 0.00 AND pf <= 0.50 THEN  
    OUTPUT 'Poor power factor'  
ELSE IF pf >= 0.51 AND pf <= 0.80 THEN  
    OUTPUT 'Fair power factor'  
ELSE IF pf >= 0.81 AND pf <= 0.95 THEN  
    OUTPUT 'Good power factor'  
ELSE IF pf >= 0.96 AND pf <= 1.00 THEN  
    OUTPUT 'Excellent power factor'  
END IF  
  
END
```

3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	pf = 0.45	Expected Output	Poor power factor
Input	pf = 0.90	Expected Output	Good power factor
Input	pf = 1.00	Expected Output	Excellent power factor
Input	pf = -0.5	Expected Output	Invalid power factor